

**Homework #8**

Due: 11:59pm, April 17

**1. [Properties of short-time CTFT]**

Recall that the short-time CTFT is defined by

$$X_c(j\Omega, \tau) = \int_{-\infty}^{\infty} x(t)w(t - \tau)e^{-j\Omega t} dt.$$

Suppose that  $\int_{-\infty}^{\infty} |w(t)|^2 dt = 1$ .

- (a) Let  $y(t) = x(t - d)$ . Derive an expression of the short-time CTFT  $Y_c(j\Omega, \tau)$  of  $y(t)$  given in terms of  $X_c(j\Omega, \tau)$ .
- (b) Let  $z(t) = x(t)e^{j\Omega_0 t}$ . Derive an expression of the short-time CTFT  $Z_c(j\Omega, \tau)$  of  $z(t)$  given in terms of  $X_c(j\Omega, \tau)$ .
- (c) Derive the following identity

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |X_c(j\Omega, \tau)|^2 d\Omega d\tau.$$

Hint: First apply Parseval's theorem for CTFT to  $x(t)w(t - \tau)$  for fixed  $\tau$ .**2. [Reconstruction condition for short-time DFT]**Let  $w[n]$  denote a finite-length window such that  $w[n] = 0$  if  $n \notin \{0, 1, \dots, L - 1\}$ . Let  $x[n]$  be a discrete-time signal. The short-time discrete Fourier transform (ST-DFT) is given by

$$X[k, l] = \sum_{n=-\infty}^{\infty} x[n]w[n - lM]e^{-j\frac{2\pi}{N}kn}, \quad l \in \mathbb{Z}, \quad k \in \{0, 1, \dots, N - 1\}.$$

Here,  $L$ ,  $M$ , and  $N$  are positive integers.

- (a) Define

$$z_l[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k, l]e^{j\frac{2\pi}{N}kn}.$$

Determine a condition on  $N$  and  $L$ , under which we have

$$x[n]w^2[n - lM] = z_l[n]w[n - lM], \quad l \in \mathbb{Z}.$$

- (b) Suppose that  $L = 2M$ . Then the windows in ST-DFT have 50% overlap. Show that

$$x[n] = \frac{1}{N} \sum_{l=-\infty}^{\infty} \sum_{k=0}^{N-1} X[k, l]w[n - lM]e^{j\frac{2\pi}{N}kn}$$

provided

$$w^2[n] + w^2[n + M] = 1, \quad n = 0, 1, \dots, M - 1.$$

- (c) Design a window  $w[n]$  from the rectangular window so that it satisfies the condition in Part (b). (Hint: Need to normalize the coefficients by dividing by an appropriate constant.)
- (d) Design a window  $w[n]$  from the Bartlett window so that it satisfies the condition in Part (b).

3. **[Continuous Haar wavelet]**

Let  $u(t)$  denote the unit step function. Let  $\psi(t) = u(t) - 2u(t - 1/2) + u(t - 1)$ . Define

$$\psi_{a,\tau}(t) = \frac{1}{\sqrt{|a|}} \psi\left(\frac{t - \tau}{a}\right).$$

- (a) Draw  $\psi_{1,0}(t)$ .
- (b) Draw  $\psi_{2^3,0}(t)$ .
- (c) Draw  $\psi_{2,k}(t)$  for  $k = 0, 1, 2$ .
- (d) Show that any distinct pair of  $\psi_{2^3,0}(t)$ ,  $\psi_{2,0}(t)$ , and  $\psi_{2,2}(t)$  has mutually orthogonal signals.

4. **[Uncertainty principle]**

Suppose that  $b(t)$  is a real-valued function satisfying

$$tb^2(t) \rightarrow 0 \quad \text{as} \quad t \rightarrow \pm\infty.$$

Define

$$E = \int_{-\infty}^{\infty} |b(t)|^2 dt,$$

and define the time and frequency means and variances by

$$\begin{aligned} \mu_t &= \frac{1}{E} \int_{-\infty}^{\infty} t |b(t)|^2 dt, \\ \sigma_t^2 &= \frac{1}{E} \int_{-\infty}^{\infty} (t - \mu_t)^2 |b(t)|^2 dt, \\ \mu_\Omega &= \frac{1}{2\pi E} \int_{-\infty}^{\infty} \Omega |B_c(j\Omega)|^2 d\Omega, \\ \sigma_\Omega^2 &= \frac{1}{2\pi E} \int_{-\infty}^{\infty} (\Omega - \mu_\Omega)^2 |B_c(j\Omega)|^2 d\Omega, \end{aligned}$$

where  $B_c(j\Omega)$  denotes the continuous-time Fourier transform (CTFT) of  $b(t)$ .

Let

$$x(t) = b(t + \mu_t),$$

so that  $x(t)$  is time centered.

**Background (Cauchy-Schwarz Inequality)** For square-integrable functions  $f(t)$  and  $g(t)$ , it holds that

$$\left| \int_{-\infty}^{\infty} g^*(t)f(t)dt \right|^2 \leq \int_{-\infty}^{\infty} |g(t)|^2 dt \cdot \int_{-\infty}^{\infty} |f(t)|^2 dt$$

with equality if and only if  $g(t) = cf(t)$  for some constant  $c$ .

(a) Use the Cauchy-Schwarz inequality to prove

$$\left| \int_{-\infty}^{\infty} tx(t) \frac{dx(t)}{dt} dt \right|^2 \leq \int_{-\infty}^{\infty} t^2 x^2(t) dt \int_{-\infty}^{\infty} \left( \frac{dx(t)}{dt} \right)^2 dt. \quad (1)$$

(b) Show that

$$E = \int_{-\infty}^{\infty} |x(t)|^2 dt \quad \text{and} \quad \mu_{\Omega} = 0.$$

(c) Using the fact that  $j\Omega X_c(j\Omega)$  is the CTFT of  $dx(t)/dt$ , show that the right-hand side of Equation (1) can be written as  $E^2 \sigma_t^2 \sigma_{\Omega}^2$ .

(d) By integration by parts, show that

$$\int_{-\infty}^{\infty} tx(t) \frac{dx(t)}{dt} dt = -\frac{1}{2}E.$$

(e) Conclude that

$$\sigma_t \sigma_{\Omega} \geq \frac{1}{2},$$

and show that equality holds if and only if

$$x(t) = Ae^{-\alpha t^2}$$

for some real  $A$  and  $\alpha > 0$ .

(f) Let

$$b(t) = \frac{1}{\sqrt{2\pi}} w(t - \tau_0) e^{j\Omega_0 t},$$

where  $w(t)$  is real-valued, even, and satisfies

$$\int_{-\infty}^{\infty} |w(t)|^2 dt = 1.$$

Show that

$$\mu_t = \tau_0, \quad \sigma_t^2 = \int_{-\infty}^{\infty} t^2 |w(t)|^2 dt,$$

and

$$\mu_{\Omega} = \Omega_0, \quad \sigma_{\Omega}^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \Omega^2 |W_c(j\Omega)|^2 d\Omega.$$

(g) Compute  $\sigma_t \sigma_{\Omega}$  when

$$w(t) = \sqrt{\frac{3}{2c}} \text{tri}(t/c)$$

for some positive constant  $c > 0$ .